

1. Database is an organised, digital collection that is stored and accessed electronically

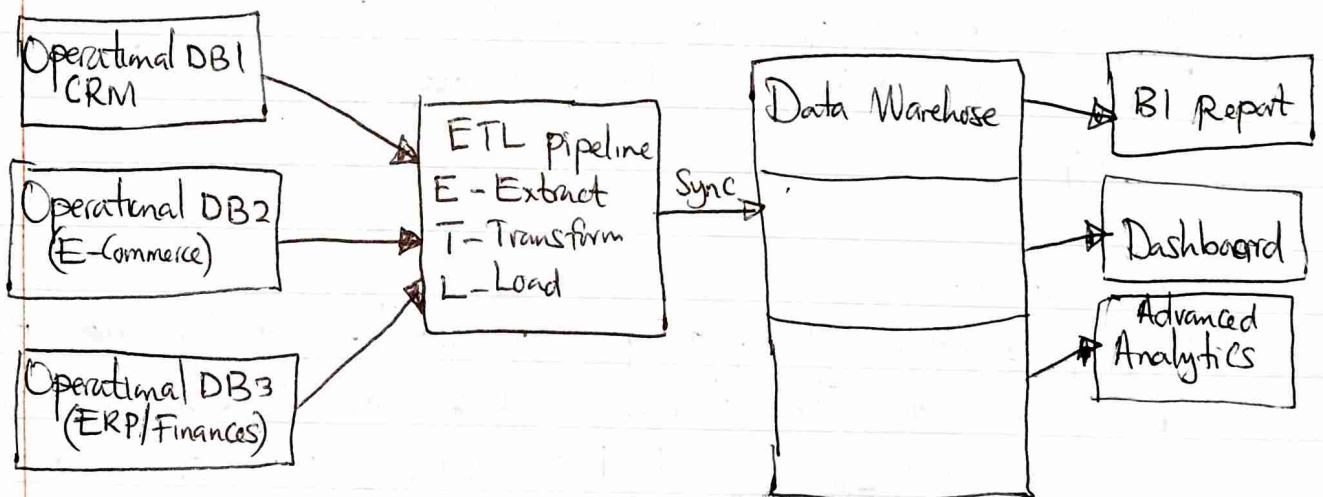
Used in Real-World example

- Streaming Service (Netflix/spotify): Database store million of videos and songs, track your watch and instantly load personalised recommendations
- Online Shopping (Amazon, takealot): Database track product prices, manage warehouse inventory level in real time and securely save your shipping and billing details

Table Customers

Customer ID	Name	City	Status
101	Bones	Pretoria	Pending
102	Chris	Makwarch	Active

2. Data Warehouse is a Central System that store, clean, organised historical data pulled from many sources databases



~~Data Warehouse~~ Regular Database (OLTP)

- ~~Run~~ Runs the business, daily transaction
- Run Small, fast queries using command like (ADD, EDIT, DELETE)
- Show Current data
- A cash register processing sales right now

Data Warehouse (OLAP)

- Analyses the business, explores historical trends
- Complex queries and reading large amount of data
- Store years history
- Pull and blends from many different system

3. Data lake is a massive, centralized repository that store vast amounts of raw data in its native, unformatted state

Data lake vs Data Warehouse

I will use analog of H_2O . Data Warehouse is like a store shelves filled with bottled water (cleaned, packaged and ready to consumed) Where Data lake is like a natural lake (raw, unfiltered water waiting to be examined or treated for various uses)

Data lake

- Raw and unrefined. Stores structured, semi-structured and unstructured
- Schema-on-Read, The structure is applied when data is pulled out
- Very low cost, designed to massive data
- Data Scientists X Engineer who want deep raw data for machine learning

Data Warehouse

- Highly Structured - Only accept cleaned, filter and formatted data
- Schema-on-Write - The structure must be defined before data goes in
- Higher Cost, Processing and structure data makes storage more ^{costly} exp
- Business Analyst and Executive, who need clean charts and quick answer

When Data lake is used

- Handling Unstructured data - When Company need to store data that does not fit into spreadsheet like customer service phone call recording, camera footage, images

- Advanced Machine learning and AI - This requires raw data
- Store everything now, Figure it out later
- Cheap for scalability

4. Data Mart is small, focused of a data warehouse built for single department

How it relates to Data Warehouse

- A data Mart usually get information directly from central data warehouse
- The Warehouse hold entire company's history and data mart filter out to show relevant for one specific business data
- It handle a small pool of data

Who Typical Uses it?

- Department Managers & Analysts
- Sale Team
- Business Executives

5 Structured Data - Organized information that's fit into rigid rows and column eg Excel Spreadsheet of Customer name, number

Semi-structured Data - Data does not fit a strict table grid but contains internal tags, markers, or hierarchies to separate the information eg JSON or XML file

Unstructured Data - Raw data, unorganised with predefined structure format eg video, audio

6 Data Modelling is the processing on creating visual map that define how data will collected, structured, stored and connected with organization system

Why it is important

- Improve System performance
- Aligns Business and IT
- Reduces Redundancy
- Prevent costly Mistakes

Two Major types of Data models

- Logical Data Model $\Rightarrow$ Structure of the data and relationship between different of information (like name, email or date)
- Physical Data Model $\Rightarrow$ Show how the data will built inside software
MYSQL, ORACLE

7. Data Mining - Process of automatically searching, sorting and analysing massive sets of raw data to discover hidden patterns, anomalies and trends which are not obvious human eyes

How it is used in Business?

Business use data mining to turn passive historical data into proactive strategies, Instead of guessing what Customer want

- Predict Customer $\Rightarrow$ Identify exactly which likely to Cancel Subscription
- Optimize Prices $\Rightarrow$ Adjust prices based on demand, weather, Competitor
- Detect Fraud - Spot ~~was~~ unusual patterns - to block fraudulent transaction

Real-World Examples

1. Retail Market Basket Analysis (eg Target & Shoprite)

Retailers mine POS receipt data to find out which items are frequently bought together

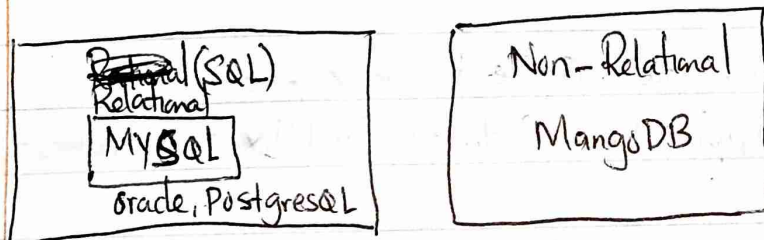
- By analysing millions of check-out records, a grocery store might discover people buy milk and eggs
- Store place these items together or create bundle promotion

2 Predictive Streaming Recommendation eg (Spotify, Netflix)

Streaming platform mine exact user behavior data every second you use their app

- The System does not look at genre of the song you like, it mines data on when you skip a track, What time of the day you listen how long time you stay on playlist
- It automatically generates highly accurate recommendation like Spotify's "Discover Weekly"

8. Database Management is the specialized software layer that act as interface between a database and end-user



1. MySQL - Relational database known for speed reliability. and easy to use

- Used - Websites and Content Management System
- Power back-end like Wordpress, Shopify

2. PostgreSQL ⇒ Object-relational ~~base~~ database praised for it's strict

- Support for highly complex queries
- Deeply Customisable and handle sophisticated data type like (Geographical Spatial data)

Used - Financial applications and geographic information System
- A company like Uber, use takealot use it for delivery service

3 Oracle Database = Relational database System, it is designed to handle gargantuan workload
- offer a strong security

Used $\Rightarrow$ Global banking network
- High volume transactional handling
- Military-grade security encryption

4 MongoDB - No-relational (NOSQL) database that stores data as flexible
- ~~is~~ JSON-like documents rather than rigid table row and column
- Also allow developers to change their data structure without breaking the database

B Used - Real time analytic, content management
- Big data Application such as Netflix or Facebook

9. ETL (Extract Transform and Load) is process extract from database transform it before load into warehouse

Extract (Gathering the Data)

Data is pulled from its original, scattered sources. These sources could be transactional database, CRM software, Web analytics, excel spreadsheet

Transform (Cleaning and Organizing the data)

The raw is sent to a temporary staging area where it is cleaned standardized and formatted so that it makes structural sense

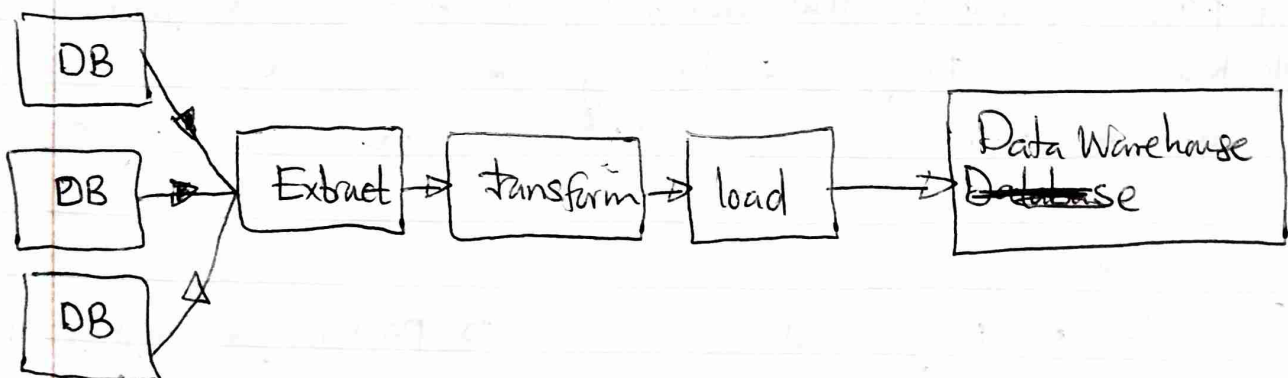
- cleaning $\Rightarrow$ Removing duplicates records, fix typos and dealing with missing data
- Standardizing $\Rightarrow$ Ensuring matching formats
- Filtering $\Rightarrow$ Sorting out useless data fields that business don't need
- Enriching $\Rightarrow$ Joining multiple table together to add context (eg matching a random Customer ID to an actual name)

Load (Storing the data)

Cleaned data and structured data is written into its final destination called Data Warehouse, available for business analyst, manager and data Scientist and view via dashboard

Why ETL is vital in Data Analytics?

- Eliminate the garbage - If an analytical report on mess, uncleaned raw data, business insight will be inaccurate. ETL act as quality Control filter
- Creates - By blending data from sales team, marketing team, and customer into one uniform format, executive can see the entire health of the company in one place
- Optimizes Query Speed - Running complex analytical equation across multiple live production apps slows them down. ETL process the heavy overnight and puts the data into warehouse
- Preserve historical data



10. CSV (Comma-Separated Values)

⇒ Plain-text, Row based where data fields are separated by comma

When to Use - for small, simple data transfers between completely different Software tool

- When you need to manually inspect data in text

- JSON (Java Script Object Notation)

- Plain text, Row based format that structure data using key-value Pairs, Support nesting data records inside other records

When to Use - for web APIs.

- Application logs
- Semi-structured data Streaming
- When Schema might change

- Apache Parquet

Highly advanced, binary Column-oriented storage format designed for the hadoop and cloud analytics ecosystem

When to Use - heavy big data analytics and data lakes

11. Artificial Intelligent

Is a field of Computer Science to create Software System capable of performing tasks that would require human intelligence. These tasks includes thing like learning from experience, recognising complex pattern, make decision and understanding natural language

Narrow AI

Is a Software programmed and trained to perform one highly specific task exceptional well, it operates under limited context and can not apply its intelligent

Real-World Example

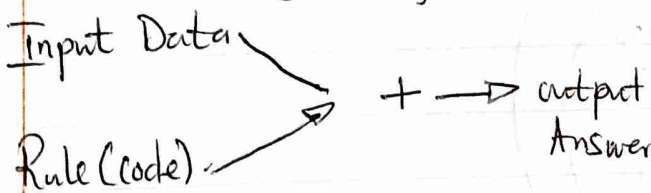
- Apple Siri, Google translate

⇒ General AI (Strong AI)

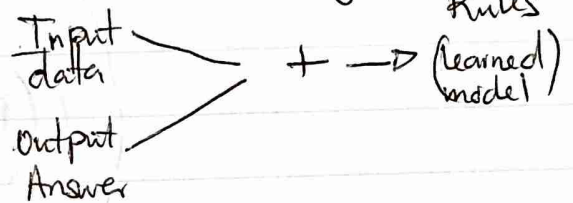
Is a form of AI where a machine possess human-level intelligent across all domain. It would have comprehensive ~~con~~ consciousness able to learn, reason, abstract and apply knowledge adaptively from one completely unrelated task to another - just like human being

12. Machine Learning is part of artificial intelligent that focuses on building software systems that can learn from data, identify pattern and make decision with minimal human intervention

Traditional Programming



Machine Learning



⇒ Software Engineer manual write code logic, Computer process those rules and process incoming data and output an answer

Formula $\text{Data} + \text{Rules (code)} \rightarrow \text{Answers}$

eg. Writing a program with strict rule if a email contain the word "crypto-lottery", move it into spam folder, If a spammer changes the wording to "crypto" it break the rules and programmer must manually update the rule

= Feed a machine learning model thousand of examples of raw data alongside the correct answers. This model analyses these examples and generates its own internal rules a (model) to predict future answers

formula. $\text{Data} + \text{Answers} = \text{Rules (ML model)}$

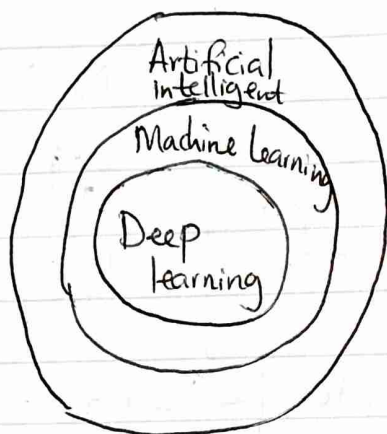
eg feed ML model 5ML real emails marked as "spam" or not spam. The system detects millions of hidden pattern (Word frequencies, sender behavior, sending times and creates its own complex rule set to block spam even if spammer change their spelling

Types of Machine Learning

- Supervised learning $\Rightarrow$ The model is trained on labelled data, every point comes with correct data eg feeding model photos of house labelled with selling prices
- Unsupervised learning - The model is given unlabeled data and must find hidden structure, grouping or patterns
- Reinforcement learning - The model learns through trial and error

13. Deep learning

is a machine learning that mimics the structure and function of the human brain to process data and create patterns for decision making



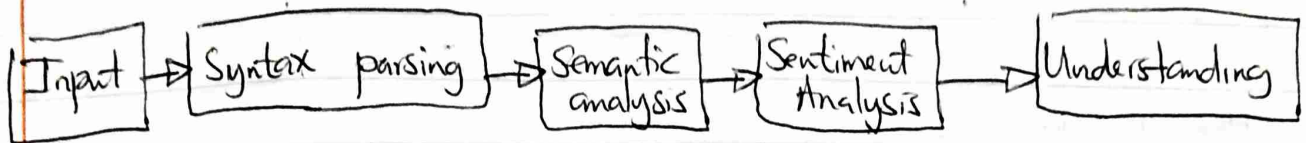
How it relates to AI and Machine learning?

1. Artificial Intelligence - Broad umbrella field of computer science dedicated to make machine smart
2. Machine learning - Where machine learn rules from data
3. Deep learning - It completely

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Natural language Processing (NLP)

Is a branch of AI that gives Computer the ability to understand, interpret, manipulate and generate human language



- Eg - Smart Assistant like Apple Siri, Amazon Alexa, Google Assistant
- Predictive Text and autocorrect like Grammar checkers
 - Email Spam filter and Sorting; Gmail use NLP to read the underlying email

Large language Model (LLM)

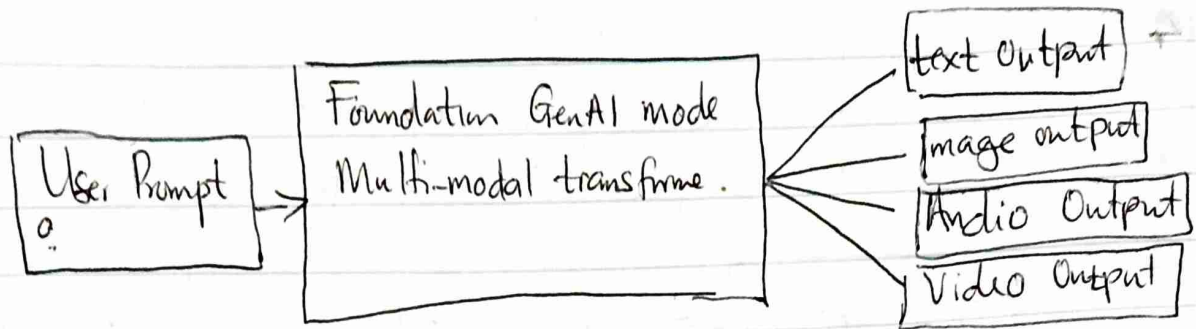
Advanced type of AI trained on massive amount of data such as books, articles and websites. It uses a neural network called transformer to learn the pattern.

How it differs from Standard Machine learning

Standard Machine learning	Large language Model
- Requires strict, labeled, task specific data table	- Pre-trained on trillions of words of unstructured web text
- Predict numbers, classes or probability	- Generates natural paragraphs, code, dialogues
- Single-task focuses. fraud detection	- Generalist Capacity, One model can summarize text
- Must completely retrained	- Adapts instantly to new tasks simply through prompting

Generative AI

AI that can create completely new content from scratch



How it Works?

- Training Phase (Pattern Recognition) is feed massive datasets (Images, books, audio clips). It breaks these inputs down into small pieces and analyze how they relate to each other
- Generation Phase (Probability Estimation) - When you type prompt, AI does not search a database. It use the mathematical probability rules it learned during training to generate content pixel and word

Popular Generative AI Tool

- ChatGPT - Use a large language Model to write essay, brainstorm ideas, debug programming code, and summarize complex business report
- Midjourney - These are text-to-image generators
- Sora - These are advanced text-to-video generators

13 Training Dataset

It is the foundation collection data such (images, text, audio clips) Used to teach AI model

Why training data quality is critical?

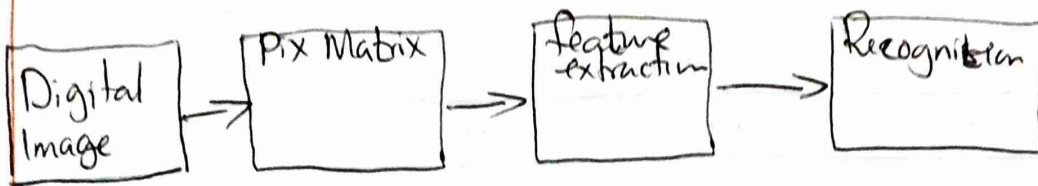
- Accurate - It can make correct prediction on brand-new data
- Reliable - It ~~per~~ perform consistently across different scenarios without crashing
- Efficient - It takes less time and computing power to train successful

What can go wrong?

- Biased data unfairness
- Incorrect data (hallucination and failure)

18 Computer Vision

AI enables computer and systems to derive meaningful information from digital images, videos and other visual input



How Machine sees and interpret images

1. Breaking images into Pixel (The input) $\Rightarrow$ To a machine, an image is just a massive grid of number.
2. Finding Patterns (Feature Extraction) $\Rightarrow$ The computer passes grids through a specialized neural network called Convolutional Neural Network (CNN)
3. Rendering a Verdict (Interpretation) $\Rightarrow$ The final layer of the network compares the extracted shapes against the statistical rules it learned during its training phase

Real-World Use Cases

- Autonomous Vehicle (Self-driving cars) $\Rightarrow$ A Tesla rely on computer vision to safely navigate through street
- Medical Imaging Analytics $\Rightarrow$ Computer vision act as highly accurate assistant for radiologist and doctors when interpreting scan.

SECTION: API, TOOL, DEVELOPER CONCEPT

19 API (Application Programming Interface)

Is a software intermediary that allows two completely different computer application to talk to each other

Restaurant Analogy

The Customer (The Client/Your app): You sit at the table looking at a menu of choices. You know what you want, but you can't go directly into the kitchen to cook it yourself.

- The Kitchen (Server/Database) = This is the backend system that hold all the ingredient raw data and cooking tool to prepare order
- The waiter (API) - The waiter is the crucial link, they take your order (Your API request) from the table, deliver it to the kitchen and bring the prepared food (API Response) back to you

HOW IT WORKS ON YOUR PHONE

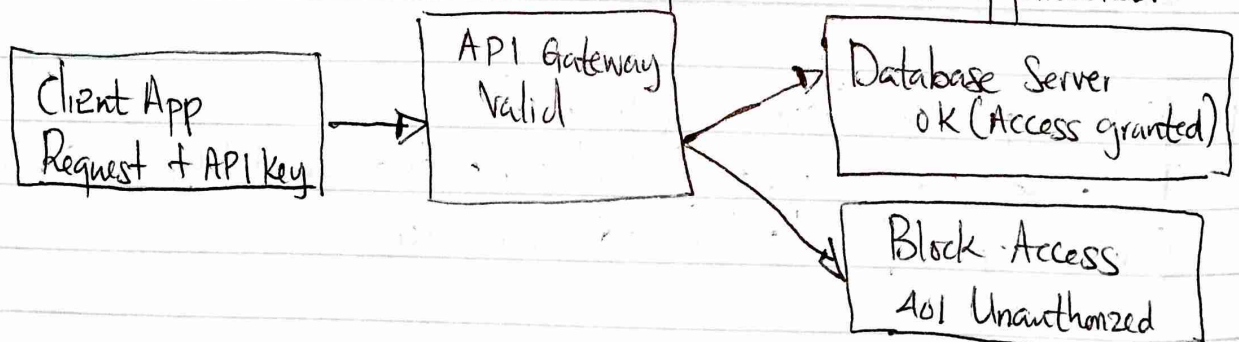
- The Map: Uber does not map the entire themselves. Instead, their app
 - Send request to the Google Maps API
- The delivery: The Google Maps API instantly fetches that data and drops it back into the Uber interface
- The payment: When you pay, Uber sends your card detail request to a payment API (bank) to process money safely

Importance of API

- Saves development time and cost
- Enable Microservices Architecture
- Protect System Security
- Power Automation and data pipeline (ETL)

20. API Key

Is a unique, long string of letters and numbers generated by a server that acts as combined username and password for an application



The Purpose of an API Key

- Authentication (Security Gatekeeper) $\Rightarrow$ It verifies incoming request is from a trusted, registered developer application and malicious hacker
- Authorization and Rate Limiting (Usage tracker) $\Rightarrow$ It allows the Servers ~~and~~ host to track, how much data an app is pulling.

How to Store and Protect API Keys

- Use environment variable (.env files) - Store keys inside a dedicated isolated environment file on your local machine or server
- Add Secret to .gitignore - If you use Git or Github to track your code always list your secret ~~and~~ environment files in a .gitignore^{internet}
- This tells your Git to never upload your secret keys to public
- Use a Secret Manager $\Rightarrow$ for production system (AWS Secret etc)
- Restricted IP and domain Scoping - Configure your API provider setting so that key only work when originate from your specific domain or server IP address

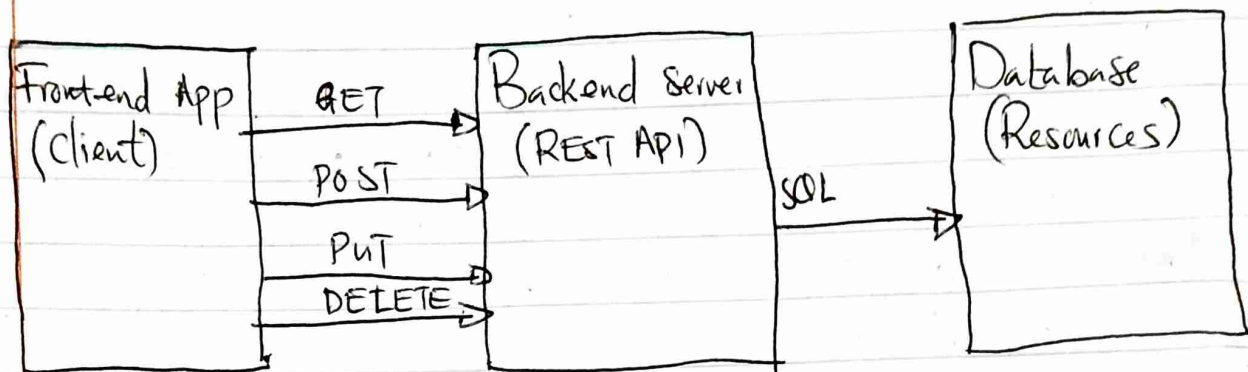
What happen if an API Key is exposed publicly?

- Massive Financial theft
- Data Breaches
- Service Deactivation & Ban

2.1 REST (Representative State Transfer)

It is an architectural style used to design web API, set of rules that different computer system to communicate over the web using existing infrastructure of the internet.

REST API HTTP workflow



4 main HTTP method in REST

- 1 GET (Read) $\Rightarrow$ Used to retrieve or view data from Server
- 2 POST (Create) $\Rightarrow$ Used to send new data to the Server
- 3 PUT (update) $\Rightarrow$ Used to update or replace an existing record
- 4 DELETE = Used to remove a specific record permanently

2.2 JSON (JavaScript Object Notation)

Is a lightweight, text-based data format used for storing and exchanging information

JSON SYNTAX structure

```
{  
  "user_id":
```

```
1042
```

Why JSON is dominate in API and data exchange?

- Extremely lightweight
- Easy for code to understand
- Human-readable
- Supports Nested Data

Example

```
{  
  "User_id": 10425,  
  "name": "Sara Jenkins",  
  "is active": true,  
  "Skills": ["Python", "SQL", "Data Analyst"],  
  "Contact_infor": {  
    "email": "Sara.j@email.com",  
    "phone": "+27 556 2161"  
  }  
}
```

23 Prompt Engineering

It is process of structure, refining and designing text inputs to ensure AI mode generate the most accurate, relevant and high quality response possible

- Why is it important skill when working with AI tool like ^{chatGPT} ~~chat~~
- Saves time with portable workflows
 - Prevent costly errors
 - Leverage platform strength
 - Protect Data Privacy

Two Real-World Example of Prompt Improvement

1 Poor prompt - Write a script to get weather data

Why it fails $\Rightarrow$ It lacks a programming language, does not specify the source of data and misses any structural requirement

Improve version $\Rightarrow$ Write a python script using requests library to fetch current temperature data from the openweathermap API for a user specified city. Include error handling for invalid city name and output the result in a clean string

EXAMPLE 2

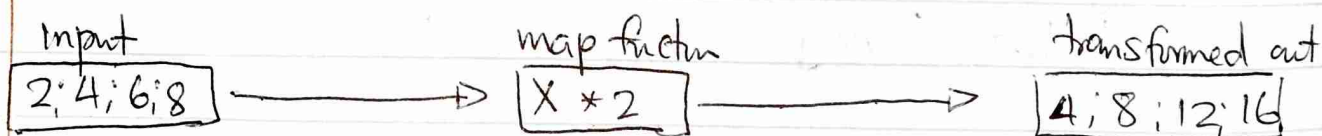
- Summarize this article for me

Why it fails $\Rightarrow$ The AI does not know the intended audience, the length constraints or the formatting preference

Improve version $\Rightarrow$ Act as professional copywriter. Summarize the following article into three bullet points. Each point must be under 15 words and written in simple language accessible to a non-technical reader

24 MAP function

Is a built-in programming tool that take a collection of data and applies a specific rule or function to every single item inside that collection, creating a brand new list with transformed results



Example

The original list of numbers (remains unchanged)

numbers = [1; 2; 3; 4; 5]

Define the transformation rule (doubling number)

def double(x):

return x * 2

Apply the map function to the list

doubled_numbers = list(map(double, numbers))

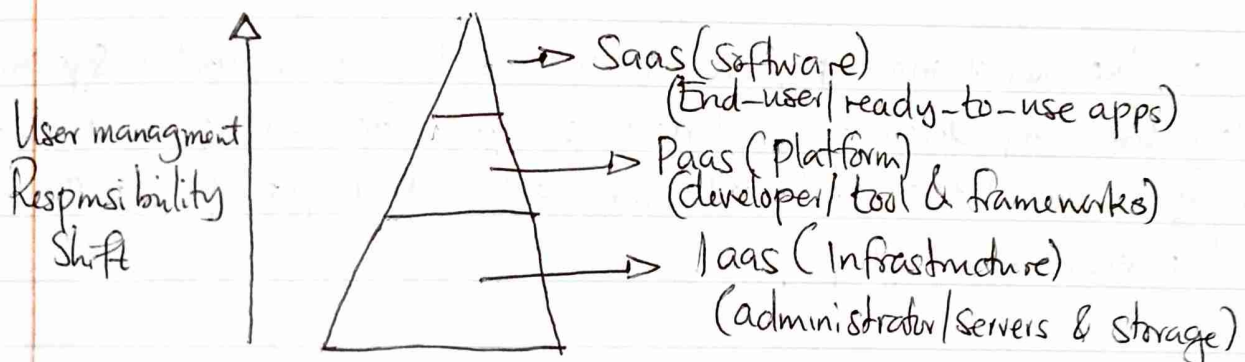
Output: [2, 4, 6, 8, 10]

Print(doubled_numbers)

2.5 Cloud Computing

Is the on-demand delivery of Computing Services - including Servers, Storage, database, networking, Software and analytics over the internet instead of buying, owning and maintain physical hard drives. You rent access to these massive infrastructure.

Cloud Service Model Pyramid (IaaS, PaaS, SaaS)



Real-World Example

→ IaaS (Infrastructure As a Service) = basic cloud tier. You rent the raw physical building blocks of computing like virtual servers, storage space and network firewalls. The cloud provider handles the actual hardware. You are completely responsible for installing and managing operating system.

Product Example

- Amazon EC2, Google Compute

PaaS (Platform as a Service) $\Rightarrow$ Manage the raw hardware and the underlying operating system and database tool.

Production example

- AWS Elastic Beanstalk

3 SaaS (Software as a Service) $\Rightarrow$ A complete, full functional software application that you access directly over the internet

Product Example

- Microsoft Office, Salesforce

26. Version Control System

Is a software tool that tracks and manage changes made to files overtime

Git is the world most popular, decentralised version control system, it widely used by developers and data analyst because prevent data loss, track who wrote what code and allow multiple people to work on the exact same file people

Two Core Git Concepts

- Commit is a digital snapshot, which means confirming before making ~~make~~ changes.
- Branch is a independent timeline, it allows you to safely test a new machine learning model or build a new ~~feet~~ feature without the working "live" production code

27 Jupyter Notebook

Is a open-source, interactive web application that allows to create and share document containing like programming code, equations visualization and explanatory text all in one place.

It is most commonly used with the PYTHON programming language

Why it is popular in Data Science

- Interactive Code execution
- Inline Visualization
- Rich text documentation
- Reproducible Workflows

28. Python

Is a high level, interpreted, general-purpose programming language known for its clear, readable syntax that resemble plain english

It has become the undisputed industry standard for Data Analytics and Artificial Intelligence

- Massive library ecosystem
- Easy to learn and Read
- Strong Community and AI support

Four Core Python libraries for data work?

- Pandas $\Rightarrow$ The ultimate data manipulation and analysis tool, Used for cleaning messy data; filtering rows; grouping metrics; merging dataset, filling in missing values

2 Numpy $\Rightarrow$ The foundational library for scientific and mathematical computing in python, fast calculation on massive data

3 Matplotlib $\Rightarrow$ A Comprehensive, low-level data visualization and charting library, Used to turn numbers into static, interactive or animated visual graphs

4 Scikit-learn $\Rightarrow$ Standard library for traditional machine learning, it provide clean, accessible tools for building predictive models, it handle task like classification eg Spam detection regression (predicting house prices), clustering (segmenting customer group)

29 Data Privacy is the principle that individual should be total control over their personal data

Why data Privacy is important?

- Prevent fraud and identity theft
- Build digital trust
- Prevent Unwanted Surveillance

Researching GDPR

GDPR (General Data Protection Regulation) is a sweeping data Privacy law enacted by the European Union that took effect on may 25, 2018. It protects anyone, it applies to any company worldwide that target or collects data from EU

7 Core principle

1. lawfulness, fairness, and transparency
2. Purpose limitation
- 3.

- 4 Accuracy
- 5 Storage Limitation
- 6 Integrity and Confidentiality
- 7 Accountability

230. Data Ethics

Is an ethics that evaluates the moral problems, rules and responsibilities associated with collecting, analyzing and using data as well as developing AI technologies. It focus on doing what is right, fair and just rather than simply what is legally permitted by code

~~Here~~ Real-World example of unethical

1. By Psychographic Profiling without Consent (The Cambridge Analytica Scandal)

A political Consulting firm ~~was~~ ^{harvested} the private data of over 87 million Facebook users without their explicit knowledge. They used data build highly specific psychological profiles to target voters with manipulate political advertisement during major election.

Why it is problematic: It involves user autonomy and informed consent. People had no idea their personality traits were being mined to actively manipulate their democratic voting choices.

2. Algorithmic Bias in Criminal Justice Systems

AI systems used by law enforcement - such as predicting policing tools or facial recognition software - have repeatedly shown much higher error rates when identifying or evaluating black and white individuals. This happens because the engineers trained the AI using historical biased arrest dataset lacking diversity.

Why it is problematic

It replicate and ~~app~~ amplifies Systematic ^{discrimination} ~~discrimina~~. When an AI tool act as an echo chamber for historical human ~~pred~~ prejudices, it results in real-world harms like the wrongful arrest of innocent individual

31. Data governance

Is a Comprehensive Collection of rules, policies, processes and data standards that define how an organization ensure data assets are accurate, secure usable, and properly managed through their entire lifecycle

Key Component of a Data Governance Framework

- Data quality Management
- Data Security and Compliance
- Data Stewardship and Roles
- Metadata Management

32 Bias in AI occurs when an AI System produces systematically prejudiced ~~Unfair~~ Unfair or inaccurate outputs.

AI does not have personal opinions or malicious intent. Instead it is feed Skewed data, it learns skewed rules, for example if a model is trained on historical data when certain group is consistently underrepresented or treated fairly. AI will assume this pattern decisions that reflect exact inequality

Real-World Example

Amazon was forced to abandon an experimental automated hiring tool after discovering it was biased against female candidates. Because the system was trained on resumes submitted over a 10-year period from historical male-dominated tech industry, AI taught itself that male applicants were preferable

Potential Consequences of AI bias

- Systematic Discrimination
- Harm in Criminal Justice
- Financial and legal liability
- Erosion of public trust

33 Data Engineers build the pipeline, data analyst look backward to explain past performance, Data Scientist look forward to predict trends, and AI Engineer build software powered by complex language and visual model.

	Data Engineer	Data Analyst	Data Scientist	AI Engineer
Primary role	Build the data plumbing (designs connects, maintains database)	Explain past trends Clean data to create charts, dashboard, report for business	Predict future outcome Uses advanced math and statistical mode to spot hidden pattern	Build smart Integrate LLM Computer vision
Core Skills	Database architecture, pipe automation, Big data Scalation	Data cleaning descriptive statistics Dash board design Business Communication	Advanced statistics Machine learning Predictive modelling	Software engineering Prompt engineering Fine-tuning API integration
tools	SQL, NoSQL Python/Scala Apache Spark AWS/Azure/GCP	Excel SQL Tableau/Power BI Basic Python/R	Python/R Jupyter Notebook Pandas/Numpy SciKit, learn	Python typescript OpenAI/LLM Langchain Vector databases
Analogy	The water engineer who build the pipes and filtration system so water flows cleanly to city	To Sommelier who tastes the filter water and explains it's qualities and history to guest	The chemist who runs a advanced experiment on water compounds to see how they behave under pressure	Inventor who build an automated smart coffee machine powered by water system

34 Business Intelligent

Is a technology driven process that analyzes raw corporate data and transforms it into actionable, easy to understand insights

Two Business intelligent tool

Microsoft Power BI - highly accessible tool that integrates seamlessly with microsoft ecosystem. It allows companies to connect to hundreds of disconnected data sources such as live sales records, marketing app

Tableau - A powerful data visualization platform known for its ability to handle massive, complex dataset and turn them into highly detailed, interactive graphics.

How THEY HELP BUSINESS

- Spotting trends and Anomalies instantly
- Tracking live KPI Metrics
- Democratizing data Access

35. The most profound shift in modern technology is transition from traditional programming to ML (and deep learning), which fundamentally flips how humans solve problem

Data Storage - Data lakes, Warehouse, ML algorithm

Computer Vision - Advanced visual recognition

Generative AI - learn the statistical probabilities of language to generate human-like text, code and art

36 ~~Am~~ Among all the advanced tech concepts covered Deep learning and Artificial Neural Network stood out as the most challenging to fully grasp initially

Resources Used to master the Concept

- 3Blue1Brown (Youtube channel)
- Tensorflow playground (Interactive Web tool)
- StatQuest with Starmer (Youtube channel)

37 Data and AI will help me combine my chemical plant experience with data-driven decision-making. I want to use Python, database, AI/ML and data analysis to identify process trends, predict potential problems, improve safety and optimize plant performance. In everyday life, AI will also help me learn faster, automate repetitive tasks, and better decisions using data.

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